

Solar System Progress Quiz

Y7 Science | 30 Marks | Lessons 1–3

Name: _____

Class: _____

Date: _____

Section A — Scale & Sizes (Lesson 1)

Q1. Name the 8 planets in order from the Sun.

[4 marks — ½ mark each]

A: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune

Q2. What is the largest planet in our solar system?

[1 mark]

A: Jupiter

Q3. What is the smallest planet in our solar system?

[1 mark]

A: Mercury

Q4. The Sun is not a planet. What type of object is it?

[1 mark]

A: A star

Q5. A student says: "Textbook diagrams show us exactly what the solar system looks like." Explain why this statement is wrong.

[2 marks]

A: Textbook diagrams are not to scale **(1)**. They make the planets look similar in size and evenly spaced, but in reality the size differences are enormous and the distances are far greater than shown **(1)**.

Q6. Explain what a scale factor is.

[2 marks]

A: A scale factor is a number you multiply real measurements by **(1)** to calculate the size or distance in a model **(1)**.

Q7. A student makes a scale model where the Sun is represented by a basketball (24 cm). The real Sun is 1,392,700 km across. Calculate the scale factor. Show your working.

[2 marks]

A: Scale factor = $24 \div 1,392,700$ **(1)** = 0.00001724 (or 1.724×10^{-5}) **(1)**

Q8. Using the scale factor from Q7, calculate the model size of Jupiter (real diameter = 142,984 km).

[2 marks]

A: $142,984 \times 0.00001724$ **(1)** = 2.47 cm (accept 2.4–2.5 cm) **(1)**

Section B — Distance & Models (Lesson 2)

Q9. Explain why you cannot show both size and distance to scale on the same model.

[2 marks]

A: If you make the planet sizes correct/visible, the distances between them would be far too large to fit (1). If you make the distances correct, the planets would be tiny invisible specks (1).

Q10. If Earth is placed 1 metre from the Sun in a scale model, approximately how far away would Neptune be?

[1 mark]

A: Approximately 30 metres

Q11. A student says: "There are lots of things between the planets — rocks, dust and gas everywhere." Using what you learned, explain why this is mostly incorrect.

[1 mark]

A: Space is overwhelmingly empty. The vast majority of the solar system is empty space/vacuum with nothing in it.

Q12. State two things that you can learn from walking out a scale model of the solar system that you cannot learn from a textbook diagram.

[2 marks]

A: Any two from: you physically experience how vast the distances are (1); you realise how much empty space there is (1); you can see that inner planets are clustered close together while outer planets are very spread out (1); you realise you can barely see the Sun marker from the outer planets (1).

Section C — Seasons & Tilt (Lesson 3)

Q13. Most people (including many adults) believe seasons are caused by Earth being closer to the Sun in summer. Explain why this is wrong.

[2 marks]

A: Earth is actually further from the Sun during Northern Hemisphere summer (1). Seasons are caused by the tilt of Earth's axis, not by distance from the Sun (1).

Q14. State the angle of Earth's axial tilt.

[1 mark]

A: 23.5°

Q15. Explain how Earth's axial tilt causes summer in the Northern Hemisphere.

[3 marks]

A: When the Northern Hemisphere is tilted towards the Sun (1), sunlight hits more directly/at a steeper angle, concentrating energy on a smaller area (1), and there are more hours of daylight giving more time for heating (1).

Q16. Describe the difference between direct and oblique sunlight hitting the ground. Explain which produces more heating and why.

[3 marks]

A: Direct sunlight hits the ground at a steep/near-vertical angle and is concentrated on a small area (1). Oblique sunlight hits at a shallow angle and is spread over a larger area (1). Direct sunlight causes more heating because the same amount of energy is concentrated on a smaller surface area, so each square metre receives more energy (1).

Mark Breakdown

Section	Topic	Marks
A	Scale & Sizes (Lesson 1)	15
B	Distance & Models (Lesson 2)	6
C	Seasons & Tilt (Lesson 3)	9
	TOTAL	30